

Chandra sai Ettneni

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DATA ENGINEER | ETL/ELT | SNOWFLAKE | AWS | AIRFLOW | AIML | PYTHON | SQL |

Python · SQL · Snowflake · AWS · Azure · Apache Airflow · ETL/ELT · Predictive Modeling · NLP · DistilBERT · MLflow

Results-driven Data Engineer with 5+ years of experience architecting production-grade **ETL/ELT pipelines**, deploying machine learning models, and delivering scalable data infrastructure across manufacturing, robotics, and enterprise technology delivering Power BI dashboards, DAX-driven KPI models, and Snowflake-based data warehouses that drive measurable ROI. Reduced pipeline runtimes and built predictive maintenance systems that cut equipment downtime by 15% and saved \$50K annually. Combines deep expertise in **Python, SQL, Snowflake, AWS, and Apache Airflow** with a consistent record of translating complex, multi-source data into business decisions that drive measurable ROI.

TECHNICAL SKILLS

- **Programming & ML/AI:** Python, SQL, Scikit-learn, XGBoost, TensorFlow, PyTorch, DistilBERT, Transformers, NLP, Fine-Tuning, Prompt Engineering, Predictive Modeling, Feature Engineering, SHAP, SMOTE, K-Means Clustering, Logistic Regression, Random Forest, Time-Series Analysis
- **Data Engineering & Analytics:** ETL/ELT Pipelines, Apache Airflow, Data Warehousing, Dimensional Modeling, Data Lineage, Data Quality Engineering, Anomaly Detection, Pandas, NumPy, PostgreSQL, MySQL, SQL Server, MLOps, UAT, Data Mining, Hadoop, Snowflake, Power BI, Tableau, DAX, Apache Spark, dbt (data build tool), Apache Kafka, Delta Lake
- **Cloud & DevOps:** AWS (SageMaker, Lambda, S3, API Gateway), Microsoft Azure ML, Google Cloud Platform (GCP), Docker, Kubernetes, CI/CD Pipelines, MLflow, Git, Terraform, Jenkins, Flask, Azure Data Factory (ADF), Power Platform (Power Apps, Power Automate)
- **Process & Collaboration:** Agile/Scrum, Stakeholder Management, Requirements Gathering, Business Requirement Documents, User Stories, JIRA, Confluence, Root Cause Analysis, Technical Documentation

PROFESSIONAL EXPERIENCE

Data Engineer — Honeywell

Jan 2025 – Present | St. Louis, MO, USA

- Architected and deployed scalable ETL/ELT pipelines ingesting REST APIs, relational databases, and flat files, increasing data availability for analytics teams by 60% while cutting pipeline onboarding time by 3 days.
- Automated end-to-end reporting workflows using Python and Apache Airflow, reducing runtime from **6 hours to 15 minutes** — a **97% reduction** — eliminating 25+ hours of weekly manual effort.
- Engineered data quality frameworks with embedded validation rules and anomaly detection, reducing downstream data quality incidents by 40% and protecting report integrity for 10+ business stakeholders.
- Administered multi-cloud data solutions across AWS, Azure, and GCP, optimizing compute and storage costs by 30% through workload-appropriate service and orchestration selection.
- Optimized SQL queries using advanced CTEs, window functions, and indexing strategies, reducing average dashboard execution time by 65% across 15+ high-traffic reports.
- Designed dimensional data models with enforced naming conventions and referential integrity, supporting 20+ downstream analytical workloads and reducing schema-related incidents by 50%.
- Built proactive monitoring and alerting frameworks with real-time failure detection and performance degradation alerts, achieving sub-10-minute mean time to detection and preventing critical data outages.
- Developed near real-time data ingestion workflows using Kafka-style event streaming concepts improving analytics freshness and reducing batch latency.
- Implemented CI/CD deployment workflows and Git-based version control practices improving pipeline release reliability and reducing deployment issues.

Senior Data Analyst — Stonedge Technologies and Robotics

Oct 2021 – Dec 2023 | India

- Constructed automated ETL pipelines for sensor data ingestion, cleaning, and transformation, saving 15+ hours of monthly manual data preparation and improving data consistency across all downstream systems.
- Delivered predictive maintenance models leveraging engineered sensor features, reducing unplanned equipment downtime by **15%** and generating **\$50,000 in annual operational cost savings**.
- Synthesized rolling window, lag, and frequency domain features from time-series sensor data, improving ML model input richness and boosting predictive accuracy by 12%.
- Applied stratified k-fold cross-validation and hyperparameter tuning to optimize Logistic Regression, Random Forest, and XGBoost models across imbalanced industrial datasets.
- Addressed class imbalance using SMOTE and class weighting, improving predictive model sensitivity by 22% and reducing false negative failure predictions driving costly reactive maintenance.
- Developed interactive Power BI dashboards with custom DAX measures, drill-through capabilities, and KPI scorecards, reducing executive reporting prep time by 8 hours/week.

- Built customer churn prediction models achieving **87% accuracy**, integrating outputs into CRM systems to directly inform retention strategies for 100,000+ customer records.
- Conducted SHAP explainability analysis to surface top churn drivers, enabling targeted marketing interventions that increased customer lifetime value by **18%**.

Data Analyst — Stoned Technologies and Robotics

Apr 2021 – Oct 2021 | India

- Performed data cleansing and validation on industrial robotics datasets, resolving missing values, duplicates, and formatting inconsistencies, improving downstream data reliability by 30%.
- Authored SQL queries for data extraction and aggregation from relational databases, reducing ad hoc reporting turnaround time by 50%.
- Contributed to Power BI dashboard development with DAX measures and custom report layouts, increasing operational metric visibility for 5+ business teams.
- Translated stakeholder requirements into accessible visualizations for non-technical users, improving data-driven decision-making adoption across departments.
- Generated summary statistics and visualizations using Python (Matplotlib, Seaborn), contributing to 3 published internal analytical reports.
- Documented data workflows and reporting processes, ensuring reproducibility and enabling team-wide knowledge sharing.

PROJECTS

AI-Enhanced Recruiting System

Python · spaCy · Scikit-learn · XGBoost · NLP · TF-IDF · Power BI

- Built an end-to-end ML recruiting pipeline to analyze 10,000+ applicant profiles, engineering NLP features including TF-IDF vectors and contextual resume signals to train an XGBoost hire-quality classifier, improving screening precision by 35%.
- Reduced manual HR review time by **40%** and surfaced candidate insights through Power BI dashboards, enabling recruiting teams to prioritize top candidates without deep technical involvement.

Production NLP Sentiment Analysis System

Python · DistilBERT · HuggingFace Transformers · AWS Lambda · API Gateway · Fine-Tuning

- Fine-tuned a DistilBERT transformer model for multi-class sentiment classification on 1,000+ daily text inputs, achieving 91% accuracy, and deployed serverlessly on AWS Lambda with sub-200ms real-time inference latency.
- Documented the system architecture in a medium article that generated 5,000+ community views and broad engagement from the data science community.

Modern Data Engineering Pipeline

Python · Airflow · Snowflake · dbt · AWS S3 · PySpark

- Built an end-to-end cloud-native ETL pipeline ingesting structured and semi-structured data into Snowflake using Apache Airflow and AWS S3.
- Developed modular dbt transformations and PySpark processing workflows handling 5M+ records with automated monitoring, logging, and alerting pipelines.
- Optimized warehouse query performance and implemented dimensional models supporting downstream BI dashboards and analytics workloads.

EDUCATION

- **Master of Science, Information Systems — GPA: 3.89/4.0 (Full-time program)**

Jan 2024 – Dec 2025

Saint Louis University | St. Louis, Missouri, USA

- **Bachelor of Technology — GPA: 3.7/4.0**

Jul 2019 – Jul 2022

JNTU | Hyderabad, India

CERTIFICATIONS

- Applied AI / Data Engineer Associate
- Gen AI Mastermind
- Python Data Analysis
- SQL for Data Analysis
- Excel for Business Analysts
- AWS Certified Data Engineer Associate (In Progress)